chapter 9

Nutrition for Hypertrophy

Proper nutrition is essential to maximizing muscle growth. This chapter focuses on the aspects of nutrition as they pertain to muscle hypertrophy; any discussion about fat loss is restricted to how it relates to the regulation of skeletal muscle mass. Moreover, the discussion is specific to healthy adults; dietary intake in those with morbidities is not addressed, nor are the implications of diet on general health and wellness.

The chapter assumes a general understanding of nutritional biochemistry. Although basic principles are presented to provide appropriate context, a detailed exploration of the nuances of the topic is beyond the scope of this book. Those interested in further exploring its intricacies are referred to the excellent resource *Advanced Nutrition and Human Metabolism*, by Gropper and Smith.

Energy Balance

Energy balance, the net difference between energy intake and energy expenditure, has a profound effect on the capacity to build muscle. Molecular signaling is altered during short-term energy deficits to favor catabolism over anabolism. Studies show that caloric restriction induces a decrease in both Akt phosphorylation and downregulation of mTOR signaling, with a corresponding activation of the FOXO family of transcription factors and upregulation of atrogin-1 and MuRF-1 expression (86, 105). Moreover, nutrient deprivation activates AMPK and NAD-dependent deacetylases, such as sirtuin 1, which in turn blunt mTOR phosphorylation (88). Because AMPK concurrently impairs translational processes while heightening high-oxidative gene expression and proteolysis, a caloric deficit would induce a high rate of protein turnover that theoretically limits increases in myofiber size (143).

Alterations in molecular signaling are consistent with research showing that caloric restriction attenuates muscle protein synthesis. Pasiakos and colleagues (103) demonstrated that postabsorptive rates of muscle protein synthesis were reduced by approximately 19% following an approximately 20% energy deficit compared to values obtained during caloric maintenance; these findings were associated with large declines in phosphorylation of Akt and 4E-BP1. Other research shows that a 5-day moderate energy deficit (~500 kcal/day) reduces muscle protein synthesis by 27% below levels attained during energy balance (8). Moreover, resistance training during the energy deficit was only sufficient to restore muscle protein synthesis levels to those seen at rest in energy balance (8). It has been speculated that the observed reductions in anabolism during times of low food availability may represent a conservation-oriented mechanism to spare ATP from unnecessary use given the energy-demanding nature of muscle protein synthesis (133). Importantly, insufficient energy intake results in an increased use of protein for fuel, regardless of protein consumption (127). That said, it is clearly possible to build muscle while losing body fat (i.e., in an energy deficit) as reported in the literature (23, 80); the extent of hypertrophy, however, will be less than that achieved in a caloric surplus.

Eucaloric conditions (i.e., an equal caloric intake and energy expenditure; also called *energy balance* or *caloric balance*) are suboptimal for inducing muscle growth as well. During periods of energy balance, the recurrent catabolism of proteins occurring in bodily organs and vital tissues is replenished in the postabsorptive state via amino acids derived predominantly from skeletal muscle (88). Although resistance training counteracts these losses, the anabolic response is nevertheless blunted, which compromises hypertrophic growth.

Alternatively, a positive energy balance alone is a potent stimulator of anabolism, even in the absence of resistance exercise training, provided that the intake of dietary protein is adequate (27). The amount of lean tissue gains associated with a combined energy surplus and resistance varies with training status. Rozenek and colleagues (119) reported that untrained subjects gained approximately 3 kg (6.6 lb) in 8 weeks when resistance training was combined with an energy surplus of approximately 2,000 kcal/day; a control group consuming a eucaloric diet did not significantly increase body mass. Virtually the entire amount of weight gain in the group consuming an energy surplus was attributed to the accretion of fat-free mass. In a study of elite athletes, Garthe and colleagues (42) randomized subjects to a diet designed to provide a surplus of approximately 500 kcal/day or an *ad libitum* intake (however much the person wants to consume). All subjects participated in the same 4-day-per-week hypertrophy-type resistance training program, which was carried out over a period of 8 to 12 weeks. Results showed a greater increase in fat-free mass in favor of those in a caloric surplus versus those at maintenance (1.7 vs. 1.2 kg, or 3.7 vs. 2.6 lb, respectively), although the results did not reach statistical significance. Interestingly, the differences in fat-free mass between the groups was specific to the lower-body musculature, where a significant advantage was noted for those in an energy surplus. Greater increases in fat-free mass associated with the energy surplus were accompanied by an increased fat deposition compared to the eucaloric condition (1.1 vs. 0.2 kg, or 2.4 vs. 0.4 lb, respectively). In a pilot study of competitive male bodybuilders, Ribeiro and colleagues (114) found that subjects consuming a high-energy diet (~6,000 kcal/day) gained more muscle mass than subjects consuming moderate amounts of energy (~4,500 kcal/day) (2.7% vs. 1.1%, respectively); however, body fat percentage increased to a substantially greater extent in the high-energy compared to the moderate-energy group (7.4% vs. 0.8%). Thus, well-trained individuals appear to use less of the surplus for lean tissue–building purposes; a higher amount goes toward adipose tissue storage. It is not clear what, if any, effect an even greater energy surplus would have had on body composition changes.

Relatively untrained individuals can benefit from a substantial energy surplus (~2,000 kcal/day); in this population, body mass gains are predominantly achieved by increasing fat-free mass at the expense of body fat, at least over the short term. In well-trained subjects, evidence suggests that a positive energy balance of 500 to 1,000 kcal/day is preferable for increasing fat-free mass (42). It has been suggested that even smaller surpluses (200 to 300 kcal/day) may be more appropriate for well-trained individuals who want to minimize body fat deposition (42). The discrepancy between populations can be attributed to the fact that untrained subjects have a higher hypertrophic potential and faster rate of growth than trained subjects, which accommodates more energy and substrate for building new tissue.

KEY POINT

To a degree, combining resistance training with a positive energy balance increases the anabolic effect; untrained people experience large gains in fat-free mass. Well-trained people use less of the energy surplus for lean tissue building and should therefore aim for a lower positive energy balance.

Macronutrient Intake

In addition to energy balance, the consumption of macronutrients (protein, carbohydrate, and lipid) is also of great importance from a nutritional standpoint. Each macronutrient is discussed in this section in terms of its relevance to muscle hypertrophy, along with practical recommendations for intake.

Protein

Dietary protein provides 4 kcal of energy per g and comprises chains of *amino acids* (nitrogenous substances containing both amino and acid groups). Over 300 amino acids have been identified in nature, but only 20 of them serve as the building blocks of bodily proteins. The anabolic effects of nutrition are primarily driven by the transfer and incorporation of amino acids obtained from dietary protein sources into bodily tissues (12). Because of variations in their side chains, the biochemical properties and functions of amino acids differ substantially (153).

Amino acids can be classified as *essential* (indispensable) or *nonessential* (dispensable). Essential amino acids (EAAs) cannot be synthesized adequately to support the body's needs and thus must be provided through the diet. Nonessential amino acids, on the other hand, can be synthesized by the body. Deprivation of even a single EAA impairs the synthesis of virtually all cellular proteins via an inhibition of the initiation phase of mRNA translation (39). Certain amino acids are classified as *conditionally essential* if they are required in the diet when amino acid use is greater than its rate of synthesis (153). Importantly, all 20 amino acids are necessary for proper cell function and growth. Table 9.1 lists the essential, nonessential, and conditionally essential amino acids.

An increase in plasma and myocellular amino acids above fasting levels initiates an anabolic response characterized by robust elevations in muscle protein synthesis. Under resting conditions this response is very transient; maximal stimulation of muscle protein synthesis occurs approximately 2 hours after amino acid ingestion and then rapidly returns to postabsorptive levels (104). Thus, muscles are receptive to the anabolic effects for a relatively short period of time in the nonexercised state.

Effect on Performance

Exercise potentiates the anabolic effect of protein intake, heightening both the magnitude and duration of the response (12). After a brief latency period, dramatic increases in muscle protein synthesis are seen between 45 and 150 minutes post-workout, and elevations are sustained for up to 4 hours in the fasted state (12). Despite this exercise-induced increase in muscle protein synthesis, post-exercise net protein balance remains negative in the absence of

TABLE 9.1 Essential, Nonessential, and Conditionally Essential Amino Acids

Essential amino acids	Nonessential amino acids
Histidine	Alanine
Isoleucine	Arginine*
Leucine	Asparagine*
Lysine	Aspartic acid
Methionine	Cysteine
Phenylalanine	Glutamic acid
Threonine	Glutamine*
Tryptophan	Glycine*
Valine	Proline*
	Serine*
	Tyrosine*

*Conditionally essential amino acids.

nutrient consumption (39). Provision of EAAs rapidly reverses this process so that protein balance becomes positive, and anabolic sensitivity is sustained for longer than 24 hours (12).

The essential amino acid leucine, one of the *branched-chain amino acids* (BCAAs), is believed to be particularly important to the regulation of muscle mass. Leucine has been shown to stimulate muscle protein synthesis both in vitro and in vivo. The mechanism of action appears to be the result of an enhanced translation initiation mediated by increased mTOR phosphorylation (104, 153). This contention is supported by findings that activation of mTOR is relatively unaffected by the other two BCAAs, valine and isoleucine (153). Leucine also has a positive effect on protein balance by attenuating muscle protein breakdown via the inhibition of autophagy (153). The influence of leucine is limited to the activation of muscle protein synthesis, not the duration; sustaining elevated muscle protein synthesis levels appears to rely on sufficient intake of the other EAAs, especially the BCAAs (107). Thus, leucine has been referred to as a nutrient "trigger" for anabolism (20).

Some researchers have proposed the concept of a *leucine threshold*, which postulates that a certain concentration of leucine in the blood must be reached to maximally trigger muscle protein synthesis (52). Research shows that a 2 g oral dose of leucine (equating to approximately 20 g of a high-quality protein such as whey or egg) is necessary to attain the threshold in young, healthy people (92), although variations in body size seemingly influence this amount. Leucine requirements are heightened in the elderly. The aging process results in desensitization of muscles to EAAs (i.e., an anabolic resistance), whereby older people require larger per-meal doses than their younger counterparts (36). Mechanistically, this is thought to be due to a dysregulation of mTORC1 signaling (see chapter 2), which in turn necessitates a higher leucinemia to trigger elevations in muscle protein synthesis (106). Katsanos and colleagues (62) found that 6.7 g of EAAs—an amount shown to be sufficient to elicit a marked anabolic response in young adults—was insufficient to elevate muscle protein synthesis above rest in an elderly group; only after supplementing the EAA bolus with 1.7 to 2.8 g of leucine did a robust increase occur. The findings suggest that older adults require approximately double the amount of leucine per serving than that of younger people to reach the leucine threshold.

It should be noted that the dose–response anabolic effects of leucine are maxed out once the threshold is attained; increasing intake beyond this point has no additional effect on muscle protein synthesis either at rest or following resistance exercise (104). Moreover, longitudinal studies in animal models have failed to show increased protein accretion from leucine supplementation in the absence of other amino acids (37, 81). This raises the possibility that supplementation of leucine alone results in an EAA imbalance that impairs transcriptional or translational function, or both. Alternatively, although leucine supplementation triggers the activation of muscle protein synthesis, the duration may not be sufficient to produce substantial synthesis of contractile elements. Either way, the findings reinforce the need for adequate consumption of the full complement of EAAs in promoting muscular development.

Requirements

The accretion of lean mass depends on meeting daily dietary protein needs. The RDA for protein is 0.8 g/kg of body mass. This recommendation is based on research showing that such an amount is sufficient for 98% of healthy, non-exercising adults to remain in a non-negative nitrogen balance. However, the RDA, although adequate for those who are largely sedentary, cannot be generalized to a resistance-trained population, particularly those who aspire to maximize muscle development. For one, the maintenance of nitrogen balance indicates that day-to-day protein losses are offset by the synthesis of new bodily proteins; gaining muscle requires a positive nitrogen balance (i.e., protein synthesis exceeds degradation over time). Moreover, intense exercise substantially increases protein turnover, heightening the need for additional substrate. In addition, the nitrogen balance technique has serious technical drawbacks that can result in lower-than-optimal protein requirements

KEY POINT

It is important to ingest protein, especially sources high in leucine, after resistance exercise to sustain muscle protein synthesis post-workout. Those seeking to maximize muscle size need substantially more protein than the RDA guidelines propose. Older adults require more protein per dose than younger adults to build appreciable muscle.

(107). Considering the totality of these factors, the protein needs of those seeking to increase muscle size are substantially higher than those listed in the RDA guidelines.

A number of studies have been carried out to determine protein requirements for those involved in resistance training. Lemon and colleagues (76) found that novice bodybuilders in the early phase of intense training required approximately 1.6 to 1.7 g/kg/day—approximately double the RDA. These findings have been confirmed by other researchers (135). Moreover, meta-analytic data comprising 49 longitudinal studies on protein supplementation combined with regimented resistance exercise reaches similar conclusions as well (95). This increased protein requirement is necessary to offset the oxidation of amino acids during exercise as well as to supply substrate for lean tissue accretion and the repair of exercise-induced muscle damage (22). The dose–response relationship between protein intake and hypertrophy appears to top out at approximately 2.2 g/kg/day (14, 22); consuming substantially larger amounts of dietary protein beyond these requirements does not elicit further increases in lean tissue mass (5). There is even some evidence that protein requirements actually decrease in well-trained lifters. Moore and colleagues (91) found that heavy resistance exercise reduced whole-body leucine turnover in previously untrained young men; an intake of approximately 1.4 g/kg/day was adequate to maintain a positive nitrogen balance over 12 weeks of training. The findings suggest that regimented resistance training causes the body to become more efficient at using available amino acids for lean tissue synthesis, thereby mitigating the need for higher protein intakes. Alternatively, hard-training bodybuilders, particularly those performing high-volume resistance training routines, seem to benefit from consuming protein at the upper end of current recommendations; given the limited research on this population, it is conceivable that requirements may even be slightly higher than those reported in the literature (115).

Recommendations for protein intake are generally based on grams per kilogram of body weight. Research studies used to derive these guidelines have been carried out in men and women with approximately 10% to 20% body fat. Extrapolating these results to reflect requirements based on fat-free mass results in values of 2.0 to 2.6 g/kg/day for men (14) and 1.8 to 2.2 g/kg/day for women (152).

Optimal total daily protein intake depends on both energy balance status and body composition. An energy surplus tends to decrease total daily protein needs because energy intake alone improves nitrogen balance, even when no protein is ingested (127). Phillips and Van Loon (107) estimated that a protein intake of up to 2.7 g/kg/day of body weight was needed during hypoenergetic periods to prevent lean tissue losses. Helms and colleagues (54) made similar recommendations, suggesting benefits for an intake of up to 3.1 g/kg/day of fat-free mass in lean, calorically restricted individuals. It has been theorized that the higher protein dosage in this population promotes phosphorylation of PBK/Akt and FOXO proteins, suppressing the proteolytic factors associated with caloric restriction and thus enhancing lean tissue preservation (88).

Quality

Protein quality also must be taken into consideration with respect to the accretion of skeletal muscle mass. The quality of a protein is primarily a function of its composition of EAAs, in terms of both quantity and proportion. A *complete protein* contains a full complement of all nine EAAs in the approximate amounts needed to support lean tissue maintenance. Alternatively, proteins low in one or more of the EAAs are considered *incomplete proteins*. With the exception of gelatin, all animal-based proteins are complete proteins. Vegetable-based proteins, on the other hand, lack sufficient amounts of various EAAs, which makes them incomplete.

Several indices are used to assess the quality of protein sources (see table 9.2). The protein digestibility–corrected amino acid score (PDCAAS) is perhaps the most widely used index; a score of 1.0 indicates that the protein is of high quality. PDCAA scores for whey, casein, and soy are all equal to 1.0, implying that there is no difference in their effects on protein accretion. Comparative studies of isolated proteins indicate that this is not the case. Wilkinson and colleagues (149) demonstrated that the post-exercise ingestion of a serving of skim milk containing 18 g of protein stimulated muscle protein synthesis to a greater extent than an isonitrogenous, isoenergetic serving of soy. Follow-up work by Tang and colleagues (134) showed that 10 g of EAAs provided by whey hydrolysate (a *fast-acting protein*) promoted markedly greater increases in mixed muscle protein synthesis after both rest and exercise compared to soy protein isolate and casein (*slow-acting proteins*). It is speculated that the fast-digesting nature of whey is responsible for this enhanced anabolic response. Theoretically, the rapid assimilation of leucine into circulation following whey consumption triggers anabolic processes to a greater extent than the slower assimilation of leucine following soy and casein consumption (107). Emerging evidence indicates the potential superiority of a blend of rapidly and slowly absorbed proteins compared to a fast-acting protein alone. Specifically, it is theorized that the addition of casein to a serving of whey results in a slower but more prolonged *aminoacidemia* (heightened amount of amino acids in the blood), which leads to higher nitrogen retention and less oxidation and therefore a prolonged muscle protein synthetic response (112). To generalize, high-quality fast-digesting proteins robustly stimulate muscle protein synthesis during the first 3 hours after consumption, whereas slow-digesting proteins exert a more graded stimulatory effect over 6 to 8 hours (34).

TABLE 9.2 Proteins and Their Respective Qualitative Scores on Commonly Used Measurement Scales

Protein source	PDCAAS	BV	PER
Casein	1.00	77	2.5
Whey	1.00	104	3.2
Egg	1.00	100	3.9
Soy	1.00	74	2.2
Beef	0.92	80	2.9
Black beans	0.75	—	—
Peanuts	0.52	—	1.8
Wheat gluten	0.25	64	0.8

PDCAAS = protein digestibility–corrected amino acid score; BV = biological value; PER = protein efficiency ratio.

Adapted from J.R. Hoffman and M.J. Falvo, 2004, "Protein—Which Is Best?" *Journal of Sports Science and Medicine* 3, no. 3 (2004): 118-130.

Caution must be exercised when attempting to draw practical conclusions from the aforementioned findings. Given that the studies measured muscle protein synthesis over short periods, they do not necessarily reflect the extended anabolic impact of protein consumption following an exercise bout. There is little evidence that consuming specific protein sources has a tangible impact on hypertrophic outcomes for those who consume adequate quantities of animal-based foods. Vegans have to be more cognizant of protein quality. Because vegetable proteins are largely incomplete, vegans must focus on eating the right combination of foods over time to ensure the adequate consumption of EAAs. For example, grains are limited in lysine and threonine, and legumes are low in methionine. Combining the two offsets the deficits, thereby helping to prevent a deficiency. Note that these foods do not have to be eaten in the same meal; they just need to be included in the diet on a regular basis.

Table 9.3 provides dietary protein intake recommendations to maximize hypertrophy.

Carbohydrate

Carbohydrates are plant-based compounds that, similar to dietary protein, also provide 4 kcal/g of energy. In broad terms, carbohydrate can be classified as either *simple* (monosaccharides and disaccharides composed of one or two

PRACTICAL APPLICATIONS

METHODS FOR ASSESSING PROTEIN QUALITY

Several methods have been developed to determine the quality of protein in a given food. These include the protein digestibility–corrected amino acid score (PDCAAS), protein efficiency ratio (PER), chemical score (CS), biological value (BV), and net protein utilization (NPU). Each method uses its own criteria for assessing protein quality, which is ultimately a function of a food's essential amino acid composition and the digestibility and bioavailability of its amino acids (122). For example, the CS method analyzes the content of each essential amino acid in a food, which is then divided by the content of the same amino acid in egg protein (considered to have a CS of 100). Somewhat similarly, the PDCAAS method is based on a comparison of the EAA content of a test protein with that of a reference EAA profile, but, as the name implies, it also takes into account the effects of digestion. The PER method takes a completely different approach; it measures weight gain in young rats that are fed a test protein as compared to every gram of consumed protein. Alternatively, both the BV and NPU methods are based on nitrogen balance: BV measures the nitrogen retained in the body and divides it by the total amount of nitrogen absorbed from dietary protein, whereas NPU simply compares the amount of protein consumed to the amount retained.

Given the inherent differences in the protein quality measured, the methods can result in large discrepancies in the reported quality of protein-containing foods. Determining which single method is the best is difficult, but a case can be made that the PDCAAS and BV methods are the most relevant to human growth because they take protein digestibility into account. That said, because each method has drawbacks, the best approach to assessing protein quality is to take multiple measures—particularly PDCAAS and BV—into account.

Recently, a new protein scoring system—the digestible indispensable amino acid score (DIAAS)—has been advocated as a superior approach to assessing protein quality. DIAAS is based on digestibility of protein in the ileum, which is believed to provide greater accuracy than current measures (108). Although DIAAS shows promise as replacement for PDCAAS and BV, many protein sources have yet to be examined using this method (108), thus compromising its practicality.

TABLE 9.3 Macronutrient Recommendations for Maximizing Hypertrophy

Macronutrient	Recommended intake
Protein	1.6-2.2 g/kg/day
Carbohydrate	≥3 g/kg/day
Dietary fat	≥1 g/kg/day ≥1.6 and 1.1 g/day* of omega-3 fatty acids for men and women, respectively

*An absolute amount, not relative to body weight.

sugar molecules, respectively) or *polysaccharides* (containing many sugar molecules). To be used by the body, carbohydrate generally must be broken down into monosaccharides, of which there are three types: glucose, fructose, and galactose. These monosaccharides are then used as immediate sources of energy or stored for future use.

Carbohydrate is not essential in the diet because the body can manufacture the glucose needed by tissues through gluconeogenesis.

Amino acids and the glycerol portion of triglycerides serve as substrate for glucose production, particularly in the absence of dietary carbohydrate. Nevertheless, there is a sound logical basis for including carbohydrate-rich foods in the diet when the goal is maximal hypertrophy.

First and foremost, as much as 80% of ATP production during moderate-repetition resistance training is derived from glycolysis (72). Substantial reductions in muscle glycogen therefore limit ATP regeneration during resistance exercise, leading to an inability to sustain muscular contractility at high force outputs. In addition, a distinct pool of glycogen is localized in close contact with key proteins involved in calcium release from the sarcoplasmic reticulum; a decrease in these stores is believed to hasten the onset of muscular fatigue via an inhibition of calcium release (100). Because of glycogen's importance as both a substrate and mediator of intracellular calcium, multiple studies have shown performance decrements in low-glycogen states. Leveritt and Abernethy (78) found that muscle glycogen depletion significantly decreased the number of repetitions performed in 3 sets of squats at 80% of 1RM. Similar impairments in anaerobic performance have been noted as a result of following a low-carbohydrate diet (74). Reduced glycogen levels also have been reported to diminish isometric strength performance (55) and augment exercise-induced muscle weakness (156). Low glycogen levels can be particularly problematic during higher-volume routines because the resulting fatigue is associated with reduced energy production from glycogenolysis (128, 147).

Effect on Performance

Although dietary carbohydrate has been shown to enhance exercise performance, only moderate amounts appear to be required to achieve beneficial effects. Mitchell and colleagues (90) found that a diet consisting of 65% carbohydrate had no greater effect on the amount of work performed during 15 sets of 15RM lower-body exercise compared to a 40% carbohydrate diet. Similarly, a low-carbohydrate diet (25% of total calories) was shown to significantly reduce time to exhaustion during supramaximal exercise, but a high-carbohydrate diet (70% of total calories) did not improve performance compared to a control diet of 50% carbohydrate (79). In contrast, Paoli and colleagues (101) reported that following a *ketogenic diet* (a diet containing less than 50 g of carbohydrate daily) for 30 days did not negatively affect anaerobic performance in a group of elite gymnasts. It is possible that these subjects became keto-adapted and therefore were better able to sustain muscular function during intense exercise. A confounding factor is that subjects in the keto group consumed substantially higher amounts of dietary protein than subjects in the control group (201 vs. 84 g, respectively). Accordingly, those in the keto group lost more body fat and retained more lean mass, which may have helped to nullify performance decrements over time.

It is less clear how longer-term reductions in carbohydrate affect markers of performance. Meirelles and Gomes (89) showed greater total-body strength improvements (combination of 8RM to 10RM testing on the leg press, triceps pushdown, and biceps pulldown) when consuming a moderately high carbohydrate diet compared to a ketogenic diet (increases of 19% vs. 14%, respectively); however, both groups were in an energy deficit throughout the study, limiting generalizability to muscle-building diets. Similar findings were reported in a cohort of CrossFit trainees; subjects who followed their customary dietary habits achieved an approximately 5 kg increase in 1RM squat strength while those following a ketogenic diet did not increase strength after 12 weeks of training (64). The caveat is that the ketogenic group was in an energy deficit whereas the control group appeared to be at caloric maintenance. Alternatively, Greene and colleagues (47) found that a 3-month ketogenic diet did not impair strength-related performance in competitive powerlifters and weightlifters compared to a higher-carbohydrate diet, despite an associated reduction in lean mass with the decreased carbohydrate intake.

Glycogen also may have a direct influence on muscle hypertrophy by mediating intracellular signaling. These actions are presumably carried out via regulatory effects on AMPK. As discussed

KEY POINT

A moderate amount of dietary carbohydrate is needed for enhancing exercise performance. It is unclear how much carbohydrate intake is needed for maximizing exercise-induced muscle hypertrophy, but 3 g/kg/day is a reasonable starting point.

in chapter 2, AMPK acts as a cellular energy sensor that facilitates energy availability. This is accomplished by inhibiting energy-consuming processes including the phosphorylation of mTORC1, as well as amplifying catabolic processes such as glycolysis, beta-oxidation, and protein degradation (46). Glycogen has been shown to suppress purified AMPK in cell-free assays (87), and glycogen depletion correlates with heightened AMPK activity in humans in vivo (151). Moreover, ketogenic diets impair mTOR signaling in rats, which is theorized to explain its antiepileptic actions (154).

Evidence suggests that low glycogen levels alter exercise-induced intracellular signaling. Creer and colleagues (29) randomized trained aerobic endurance athletes to perform 3 sets of 10 repetitions of knee extensions with a load equating to 70% of 1RM after following either a low-carbohydrate diet (2% of total calories) or a high-carbohydrate diet (77% of total calories). Muscle glycogen content was markedly lower in the low- compared to high-carbohydrate condition (~174 vs. ~591 mmol/kg dry weight). Early-phase Akt phosphorylation was significantly elevated only in the presence of high glycogen stores; phosphorylation of mTOR mimicked the Akt response, although the ERK1/2 pathway was relatively unaffected by muscle glycogen content status. Glycogen inhibition also has been shown to impede $p70^{S6K}$ activation, inhibit translation, and decrease the number of mRNA of genes responsible for regulating muscle growth (26, 31). Conversely, Camera and colleagues (21) reported that glycogen levels had no effect on anabolic signaling or muscle protein synthetic responses during the early post-workout recovery period following performance of a multiset lower-body resistance training protocol. A plausible explanation for contradictions between studies is not readily apparent.

Research also shows that carbohydrate intake influences hormone production. Testosterone concentrations were consistently higher in healthy males following 10 days of high-carbohydrate compared to low-carbohydrate consumption (468 vs. 371 ng/dL, respectively), despite the fact that the diets were equal in total calories and fat (4). These changes were paralleled by lower cortisol concentrations in high- versus low-carbohydrate intake. Similar findings are seen when carbohydrate restriction is combined with vigorous exercise. Lane and colleagues (73) reported significant decreases of over 40% in the free-testosterone-to-cortisol ratio in a group of athletes consuming 30% of calories from carbohydrate following 3 consecutive days of intense training; no alterations were seen in a comparative group of athletes who consumed 60% of total calories as carbohydrate. Whether such alterations in hormone production negatively affect muscular adaptations is unknown.

Although a majority of research on the ketogenic diet has been carried out in sedentary individuals, emerging data are beginning to shed light on the longitudinal effects of low-carbohydrate versus carbohydrate-rich diets on resistance training–induced hypertrophic adaptations. Early research in overweight, untrained women showed that adoption of a ketogenic diet combined with resistance exercise did not increase lean mass after 10 weeks, whereas a control group following their usual and customary diet achieved an increase of 1.6 kg (3.5 lb) of lean mass (60). Vargas and colleagues (144) randomized resistance-trained men to perform an 8-week resistance training program while consuming either a ketogenic diet (~42 g carbohydrate/day) or nonketogenic diet (~55% of total calories from carbohydrate); protein intake was equated between conditions (2 g/kg/day) and both groups were individually supervised by a nutritionist. Results showed that the nonketogenic group gained 1.3 kg (2.9 lb) of lean mass while the ketogenic diet group showed a slight decrease. These findings are consistent with those of Kephart and colleagues

(64), who showed that CrossFit trainees following their usual and customary diets modestly increased vastus lateralis thickness and leg lean mass while those following a ketogenic diet had decreases in both measures. Meirelles and Gomes (89) also showed greater hypertrophic improvements in the quadriceps when consuming a moderately high carbohydrate versus a ketogenic diet (4.0% vs. –2.1%, respectively); however, changes in upper-arm mass were fairly similar between conditions. In perhaps the most relevant study of well-trained individuals, competitive powerlifters and weightlifters lost 2.3 kg (5.1 lb) of lean mass when following a ketogenic versus a standard carbohydrate diet undertaken in crossover fashion (47). Collectively, findings in the current literature indicate that very low carbohydrate diets are suboptimal for maximizing muscle growth. However, it is important to note that the ketogenic diet groups in these studies were most likely in a caloric deficit, which in itself impairs anabolic responses to resistance training. It therefore is difficult to determine whether detrimental effects are caused by a severe reduction in carbohydrate or a low energy intake, or a combination of both.

Requirements

Based on current evidence, no definitive conclusions can be made for ideal carbohydrate intake from the standpoint of maximizing hypertrophic gains. Slater and Phillips (128) proposed an intake of 4 to 7 g/kg/day for strength-type athletes, including bodybuilders. Although this recommendation is reasonable, its basis is somewhat arbitrary and does not take into account large interindividual variations with respect to dietary response. The use of carbohydrate as a fuel source both at rest and during exercise of various intensities varies by as much as 4-fold among athletes; it is influenced by a diverse array of factors, including muscle fiber composition, diet, age, training, glycogen levels, and genetics (53). At the very least, it would be prudent to consume enough carbohydrate to maintain fully stocked glycogen stores. The amount needed to accomplish this task varies based on several factors (e.g., body size, source of carbohydrate, volume of exercise), but a minimum intake of approximately 3 g/kg/day seems to be sufficient. Additional carbohydrate intake should then be considered in the context of individual preference and response to training.

Table 9.3 provides the recommended intake of carbohydrate to maximize hypertrophy.

Dietary Fat

Fat, also known as *lipid*, is an essential nutrient that plays a vital role in many bodily functions. These functions include cushioning the internal organs for protection; aiding in the absorption of vitamins; and facilitating the production of cell membranes, hormones, and prostaglandins. At 9 kcal/g, fat provides more than twice the energy per unit as protein or carbohydrate.

Dietary fat is classified into two basic categories: *saturated fatty acids* (SFAs), which have a hydrogen atom on both sides of every carbon atom (i.e., the carbons are saturated with hydrogens), and *unsaturated fatty acids,* which contain one or more double bonds in their carbon chain (i.e., a missing hydrogen atom along the carbon chain). Fats with one double bond are called *monounsaturated fatty acids* (MUFAs), of which oleate is the most common. Fats with two or more double bonds are called *polyunsaturated fatty acids* (PUFAs). There are two primary classes of PUFAs: omega-6 linoleate (also called *omega-6* or n*-6 fatty acids*) and omega-3 alpha-linolenate (also called *omega-3* or n*-3 fatty acids*). Because of an absence of certain enzymes, these fats cannot be manufactured by the human body and are therefore an essential component in food.

Further subclassification of fats can be made based on the length of their carbon chains. The chains range between 4 and 24 carbon atoms, and hydrogen atoms surround the carbon atoms. Fatty acids with chains of 4 to 6 carbons are called *short-chain fatty acids*; those with chains of 8 to 12 carbons are called *medium-chain fatty acids,* and those with more than 12 carbons are called *long-chain fatty acids*.

Effect on Performance

Dietary fat consumption has little if any effect on resistance performance. As previously noted, resistance training derives energy primarily from anaerobic processes. Glycolysis, particularly fast glycolysis, is the primary energy system driving moderate-repetition, multiset protocols (72). Although intramuscular triglyceride does provide an additional fuel source during heavy resistance training (38), the contribution of fat is not a limiting factor in anaerobic exercise capacity.

Fat consumption has been shown to have an impact on testosterone concentrations. Testosterone is derived from cholesterol, a lipid. Accordingly, low-fat diets are associated with a modest reduction in testosterone production (35, 50). The relationship between dietary fat and hormone production is complex, however, and is interrelated with energy intake, macronutrient ratios, and perhaps even the types of dietary fats consumed (145). Moreover, very high-fat meals actually have been shown to suppress testosterone concentrations (146). There appears to be an upper and lower threshold for dietary fat intake to optimize testosterone production, above or below which hormone production may be impaired (121). What, if any, effect these modest alterations in testosterone levels within a normal physiological range have on hypertrophy remains uncertain at this time.

Evidence shows that the type of dietary fat consumed has a direct influence on body composition. Rosqvist and colleagues (117) demonstrated that overfeeding young men and women of normal weight with foods high in *n*-6 fatty acids caused an approximately 3-fold increase in lean tissue mass compared to comparable overfeeding with saturated fats. It is conceivable that results were related to differential effects on cell membrane fluidity between the types of fats consumed. Specifically, PUFAs have been shown to enhance the fluidity of the membrane, whereas SFAs have the opposite effect (96). Cell membranes serve a critical role in regulating the passage of nutrients, hormones, and chemical signals into and out of cells. When membranes harden, they are desensitized to external stimuli, inhibiting cellular processes including protein synthesis. Alternatively, cell membranes that are more fluid have an increased permeability, allowing substances and secondary messenger molecules

KEY POINT

Polyunsaturated fatty acids (PUFAs) are conceivably important for enhancing muscle protein synthesis and should be prioritized over saturated fatty acids (SFAs). A minimum of 1 g/kg/day of dietary fat appears sufficient to prevent negative hormonal alterations.

associated with protein synthesis to readily penetrate the cytoplasm (131). This provides a physiological basis for a beneficial impact of PUFAs on muscle protein synthesis, compared to the negative effects of excess SFAs, which reduce the fluidity of the cell membrane (18).

The *n*-3 fatty acids are believed to have a particularly important role in protein metabolism. A number of studies show that *n*-3 fatty acid supplementation results in greater accretion of muscle proteins compared to other types of fats in both animals (15, 44) and humans (97, 120, 129). These effects may be in part regulated by *n*-3 fatty acid–mediated increases in cell membrane fluidity (3), which facilitates an enhanced mTOR/p70^{S6K} signaling response (129). Additional benefits may be attributed to reductions in protein breakdown associated with the inhibition of the ubiquitin–proteasome pathway (148), which theoretically would lead to a greater accretion of muscle proteins. Although these findings are intriguing, the aforementioned studies were not carried out in conjunction with a structured resistance training protocol; limited research into combining *n*-3 supplementation with regimented exercise shows conflicting results (118). It therefore remains speculative as to what, if any, effects *n*-3 fatty acids have for those seeking to maximize hypertrophic adaptations.

Requirements

Similar to carbohydrate intake, no concrete guidelines can be given as to the amount of dietary fat needed to maximize muscle growth. As a general rule, fat intake should comprise the balance of calories after accounting for the consumption of protein and carbohydrate. Given a caloric surplus, there is no problem meeting basic needs for dietary lipids. Based on limited data, a minimum of 1 g/kg/day appears sufficient to prevent hormonal alterations. It seems prudent to focus on obtaining the majority of fat calories from unsaturated sources. The PUFAs, in particular, are essential not only to proper biological function, but seemingly to maximize muscle protein accretion as well.

Recommendations for dietary fat intake to maximize hypertrophy are shown in table 9.3.

Feeding Frequency

The frequency of nutrient consumption can influence muscle protein accretion. Given evidence of a leucine threshold, a case can be made for consuming multiple protein-rich meals throughout the day. Studies show dose-dependent and saturable effects at 10 g of EAAs, which is equivalent to approximately 20 g of a high-quality protein source (12). This is consistent with the "muscle full" concept, which proposes that muscle protein synthesis becomes unresponsive to further increases in intake once the saturable level is reached (11). Circulating amino acids are then shunted to fuel other protein-requiring processes, to suppress proteolysis, or toward oxidation (33). With muscle full status, myofibrillar muscle protein synthesis is stimulated within 1 hour, but the stimulation returns to baseline within 3 hours despite sustained elevations in amino acid availability (34). Hence, it is hypothesized that consuming protein every few hours throughout the day optimizes muscle protein accretion by continually elevating levels of muscle protein synthesis and attenuating muscle protein breakdown (13, 128).

Support for frequent feedings was provided by Areta and colleagues (7), who investigated the effects of various distributions of protein consumption on anabolic responses.

KEY POINT

It is hypothesized that consuming protein every few hours throughout the day optimizes muscle protein accretion by continually elevating levels of muscle protein synthesis and attenuating muscle protein breakdown.

Twenty-four resistance-trained men were randomized to consume 80 g of whey protein as either a pulse feeding (8 × 10 g every 1.5 hours), an intermediate feeding (4 × 20 g every 3 hours), or a bolus feeding (2 × 40 g every 6 hours) during 12 hours of recovery after a resistance training bout. Results showed that the intermediate feeding condition was superior to either the pulse or bolus feeding condition for stimulating muscle protein synthesis over the recovery period. The findings are consistent with the leucine threshold concept. The 20 g of whey provided in the intermittent feeding condition was sufficient to hit the threshold, and more frequent feedings at this saturable amount seemingly kept muscle protein synthesis elevated throughout the day. Alternatively, the pulse feeding of 10 g was insufficient to trigger leucine's maximal effects, whereas the bolus feeding was not provided frequently enough to sustain muscle protein synthesis elevations. Several issues with this study hinder the ability to extrapolate findings in practice. Although the provision of only a fast-acting protein (whey) provides the necessary control to tease out confounding effects from other nutrients, it has little relevance to real-life eating patterns. Consumption of a mixed meal increases transit time through the gut, which would necessarily require higher protein intakes to provide a leucine trigger and then release the remaining amino acids slowly over the succeeding 5 hours. Moreover, the 80 g dose of total daily protein provided to resistance-trained young male subjects is far below that needed to maintain a non-negative protein balance.

A recent study by Mamerow and colleagues (85) provides additional insight into the topic. In a randomized crossover design, 8 healthy subjects followed isoenergetic and isonitrogenous diets at breakfast, lunch, and dinner for two separate 7-day periods. During one condition, protein was distributed approximately evenly throughout each meal; in the other, it was skewed so that almost 2/3 of the daily protein dose was consumed at dinner. Protein intake was sufficient for maximal anabolism, amounting to 1.6 g/kg/day. All meals were individually prepared by the research staff. Consistent with the findings of Areta and colleagues, results showed that muscle protein synthesis was approximately 25% greater when protein intake was evenly distributed compared to a skewed distribution.

Several longitudinal studies have investigated the effects of protein intake frequency on body composition in conjunction with mixed meals. In a 2-week intervention on elderly women, Arnal and colleagues (9) demonstrated that protein pulse feeding (women consumed 79% of total daily protein in a single feeding of ~52 g) resulted in a greater retention of fat-free mass compared to a condition in which protein feedings were equally spread over the course of four daily meals. Alternatively, a follow-up study by the same researchers using an almost identical nutritional protocol found no difference between pulse- and spread-feeding frequencies in a group of young women (10). These findings are consistent with those of Adechian and colleagues (2), who reported no differences in body composition between protein pulse feeding (80% protein in one meal) and spread feeding (four equally spaced portions of protein) in a group of young obese women. The discrepancies in studies can seemingly be attributed to the age-related differences in the subjects. As previously mentioned, the aging process desensitizes muscle to protein feedings, resulting in a greater per-meal requirement to hit the leucine threshold. It is estimated that elderly people require high-quality protein in a dose of approximately 40 g for a maximal anabolic trigger; younger people require approximately half this amount (150, 155). The spread-feeding group in the study of elderly subjects consumed approximately 26 g of protein per meal (9), which would put them far below the leucine threshold during each feeding. The pulse-feeding group, on the other hand, would have hit the leucine threshold in the 80% protein meal, which may have been sufficient to promote a superior anabolic effect. In the studies of young subjects (2, 10), the spread-feeding group consumed >20 g per serving, thus exceeding the theoretical leucine threshold. A limitation of these studies is that subjects did not perform resistance exercise, thereby impeding generalizability to those seeking to maximize hypertrophy.

Research from Grant Tinsley's lab on intermittent fasting protocols provides further insights into the topic. In the first of their studies (137), untrained recreationally active men were randomized to either a control group that consumed their normal diet or an experimental group that consumed all daily calories within a 4-hour period on nonworkout days (4 days per week) with no restrictions on the training days. Both groups performed a regimented bodybuilding-style resistance training program 3 days per week. At the conclusion of the 8-week study period, greater gains in lean soft-tissue mass were seen in the control diet, indicating that restricting feedings to 4 hours impaired anabolism. However, follow-up studies in resistance-trained men (93) and women (138) showed similar increases in lean mass and other measures of hypertrophy when the time-restricted feeding groups consumed all their daily food in an 8-hour window rather than spreading out consumption across the day and evening. These findings suggest that the body becomes more efficient in using larger boluses of protein for tissue-building purposes when nutrient consumption is restricted to short daily time frames, at least over an 8-hour feeding window. That said, the findings are preliminary and must be considered in the context of nutritional self-reporting, which historically is inaccurate (123). Given that the anabolic effect of a protein-rich meal lasts 5 to 6 hours (75), it seems prudent that people seeking to maximize hypertrophy should aim for a protein intake of 0.4 to 0.55 g/kg/meal spread across at least four meals to consume 1.6 to 2.2 g/kg/day (126). Increasing protein distribution across more than four daily meals is an option for those who prefer more frequent feedings, but no additional benefits appear to be derived from the approach (83).

The consumption of protein immediately before bedtime has been proposed as a strategy to enhance anabolism (130). Recommendations generally advise using casein for a protein source because it is slow acting and therefore released over the duration of sleep. While research does show enhanced anabolism from presleep supplementation, the benefits appear to be derived from meeting total daily protein needs rather

PRACTICAL APPLICATIONS

HOW MUCH PROTEIN CAN THE BODY USE FOR MUSCLE-BUILDING IN A SINGLE MEAL?

A common claim heard in fitness circles is that the body can absorb only 20 to 30 g of protein in a single feeding. This belief is often used in support of eating protein-rich meals every few hours throughout the day. However, the veracity of such claims are dubious.

First and foremost, it is important to differentiate the term *absorption* from *utilization*. From a nutritional standpoint, absorption refers to the passage of nutrients from the gut into circulation. As such, there is virtually no limit to how much protein a person can absorb from a given meal. After digestion, the protein's constituent amino acids traverse the intestinal wall and enter circulation where virtually all become available for use at the tissue level. A potential issue occurs when a person ingests individual free-form amino acids, which can bring about competition for transport through the enterocytes. In this case, amino acids present in the highest concentrations are preferentially absorbed at the expense of those in lesser concentrations (49).

With this information in mind, the more pertinent question is how much protein the body can use from a single feeding to build muscle. This has important implications for optimizing muscle development because amino acids not used are either oxidized for energy or transaminated to form alternative bodily compounds (94).

Research by Areta and colleagues (7) indicates only a limited amount of protein can be used at the tissue level. The study randomized subjects to perform 4 sets of leg extension exercise and then consume post-exercise protein at rest under the following conditions: 8 servings of 10 g every 1.5 hours, 4 servings of 20 g every 3 hours, or 2 servings of 40 g every 6 hours. Results showed that the 20 g dose had the greatest effect on muscle protein synthesis over a 12-hour recovery period, suggesting the upper threshold for use is less than 40 g. Although these results may seem compelling on the surface, several confounding variables must be considered when drawing practical inferences from the data. For one, the researchers used whey as the protein source. Whey is a fast-acting protein. Its absorption rate is estimated to be approximately 10 g per hour (16). This implies that the 40 g dose would have been completely absorbed within 4 hours, long before subjects in this group consumed their subsequent dose at the 6-hour mark. In contrast, the absorption rate for cooked egg protein is approximately 3 g an hour (16), resulting in a much more prolonged anabolic effect. Moreover, people most often consume protein in the context of whole foods that also contain combinations of carbohydrate and fat; the inclusion of these additional nutrients further slows absorption, allowing a more time-released entry of the amino acids into circulation. Finally, subjects were young males with an average body weight of approximately 82 kg (181 lb); the total protein intake of 80 g therefore was far below their daily requirement to maximize anabolism (approximately 130 to 180 g). In sum, each of these factors, alone or in combination, may have unduly influenced findings and thus limit extrapolation to real-world scenarios (126).

A subsequent study by Macnaughton and colleagues (84) indicates that the type of exercise program also may have been a confounding variable. In this study, subjects engaged in a total-body resistance training program, as opposed to the study by Areta and colleagues (7), which included just leg extension exercise. Immediately post-exercise, subjects received either a 20 or 40 g dose of whey protein. In contrast to the findings of Areta and colleagues (7), the myofibrillar fractional synthetic rate was approximately 20% greater when consuming the 40 g versus 20 g protein dose. This suggests that activating a larger amount of muscle mass increases the body's ability to use higher amounts of protein for tissue building.

More recently, a study in older adults demonstrated a clear dose–response relationship between the amount of protein consumed in a single bolus and measures of muscle protein synthesis following a bout of total-body resistance exercise (57). Protein synthetic rates progressively increased with consumption of 0, 15, 30, and 45 g, with higher doses showing statistically greater effects than the lower doses. Thus, elderly individuals may require an even higher post-workout protein dose to achieve a comparable level of anabolism to that of younger individuals.

Data from intermittent fasting research provides longitudinal evidence that per-dose utilization may be even higher than otherwise thought. Tinsley and colleagues (138) showed a similar accretion of fat-free mass and skeletal muscle hypertrophy in trained women over the course of a supervised 8-week total-body resistance exercise program, regardless of whether food was consumed throughout the day or restricted to an 8-hour window. Although a mechanistic rationale has yet to be determined for the findings, it can be speculated that perhaps the body becomes more efficient at using protein for tissue-building purposes when feeding is restricted within limited time boundaries, sparing oxidation of amino acids.

In summary, there is little doubt that a threshold exists for how much protein an individual can utilize in a given meal; beyond a certain dose, amino acids will be oxidized for energy rather than used for muscle building. However, evidence indicates that the threshold appears to be higher than the common claim of 20 to 30 g in a sitting. It is important to note that several external factors influence the threshold, including the protein source, the co-consumption of other nutrients, and the amount of muscle involved in the exercise bout. Individual factors such as age, training status, and the amount of lean body mass must be considered as well.

than from the timing of consumption (61). Thus, the strategy can be employed as a means to ensure that daily protein targets are met, but results are not dependent on intake before sleep.

Nutrient Timing

Nutrient timing is a strategy to optimize the adaptive response to exercise. The post-exercise period is often considered the most critical part of nutrient timing from a muscle-building standpoint. This is based on the premise of an *anabolic window of opportunity,* whereby the provision of nutrients within approximately 1 hour of the completion of exercise enhances the hypertrophic response to the bout (65). According to nutrient timing theory, delaying consumption outside of this limited window has negative repercussions on muscle growth. Some researchers have even postulated that the timing of nutrient consumption is of greater importance to body composition than absolute daily nutrient consumption (24).

Protein is clearly the critical nutrient for optimizing the hypertrophic response. As previously noted, anabolism is primarily mediated by EAAs, with minimal contribution from nonessential amino acids (19, 140). It has been proposed that consumption of carbohydrate potentiates the anabolic effects of post-exercise protein intake, thereby increasing muscle protein accretion (58).

PRACTICAL APPLICATIONS

EATING FREQUENCY FOR HYPERTROPHY

Given that the anabolic effect of a protein-rich meal lasts 5 to 6 hours (75), it would be prudent for people seeking to maximize hypertrophy to spread protein intake of 0.4 to 0.55 g/kg/meal across at least four meals to reach a *minimum* of 1.6 to 2.2 g/kg/day (126). This frequency pattern ensures that the body remains in anabolism over the course of the day and takes full advantage of the >24-hour sensitizing effect of resistance training on skeletal muscle (12).

The rationale for nutrient timing is well-founded. Intense exercise causes the depletion of a substantial proportion of stored fuels (including glycogen and amino acids) and elicits *structural perturbations* (disruption or damage) of muscle fibers. Hypothetically, providing the body with nutrients following such exercise not only facilitates the repletion of energy reserves and remodeling of damaged tissue, but does so in a supercompensated manner that ultimately heightens muscular development. Indeed, numerous studies support the efficacy of nutrient timing for acutely increasing muscle protein synthesis following a resistance training bout over and above that of placebo (111, 139, 141, 142). These findings provide compelling evidence that exercise sensitizes muscles to nutrient administration.

Anabolic Window of Opportunity

The concept of an anabolic window of opportunity was initially formulated from acute muscle protein synthesis data. In one of the earliest studies on the topic, Okamura and colleagues (99) found a significantly greater protein synthetic response when dogs were infused with amino acids immediately after 150 minutes of treadmill exercise compared to delaying administration for 2 hours. Subsequently, a human trial by Levenhagen and colleagues (77) showed that lower-body (and whole-body) protein synthesis increased significantly more when protein was ingested immediately following 60 minutes of cycling at 60% of $\dot{V}O_2max$ versus delaying consumption by 3 hours. A confounding issue with these studies is that both involved moderate-intensity, long-duration aerobic exercise. This raises the possibility that results were attributed to greater mitochondrial and perhaps other sarcoplasmic protein fractions as opposed to the synthesis of contractile elements. In contrast, Rasmussen and colleagues (111) investigated the acute impact of protein timing after resistance training and found no significant differences in the protein synthetic response between consuming nutrients 1 hour versus 3 hours post-exercise.

The aforementioned studies, although providing interesting mechanistic insight into post-exercise nutritional responses, are limited to generating hypotheses regarding hypertrophic adaptations as opposed to drawing practical conclusions about the efficacy of nutrient timing for building muscle. Acute measures of muscle protein synthesis taken in the post-workout period are often decoupled from the chronic upregulation of causative myogenic signals (28) and do not necessarily predict long-term hypertrophic adaptations from regimented resistance training (136). In addition, post-workout elevations in muscle protein synthesis in untrained subjects are not replicated in those who are resistance trained (1). The only way to determine whether a nutrient's timing produces a true hypertrophic effect is by performing training studies that measure changes in muscle size over time.

Effect of Post-exercise Protein on Hypertrophy

A number of longitudinal studies have directly investigated the effects of post-exercise protein ingestion on muscle growth. The results of these trials are contradictory, seemingly because of disparities in study design and methodology. In an attempt to achieve clarity on the topic, my lab conducted a meta-analysis of the protein-timing literature (124). Inclusion criteria were that the studies had to involve randomized controlled trials in which one group received protein within 1 hour pre- or post-workout and the other did not for at least 2 hours after the exercise bout. Moreover, studies had to span at least 6 weeks and provide a minimum dose of 6 g of EAAs—an amount shown to produce a robust increase in muscle protein synthesis following resistance training (19, 65). Twenty-three studies were analyzed comprising 525 subjects. Simple pooled analysis of data showed a small but statistically significant effect (0.20) on muscle hypertrophy favoring timed protein consumption. However, regression analysis found that virtually the entire effect was explained by greater protein consumption in the timing group versus the nontiming group (~1.7 g/kg vs. 1.3 g/kg, respectively). In other words, the average protein consumption in the nontimed groups was well below what is deemed necessary for maximizing the protein synthesis associated with resistance training. Only a few studies actually endeavored to match protein intake between conditions. A subanalysis of these studies revealed no statistically significant effects associated with protein timing. The findings provide strong evidence that any effect of protein timing on muscle hypertrophy is relatively small, if there is one at all. That said, there is no discernable detriment to consuming protein close to a training bout and, given that even relatively modest effects may be practically meaningful, the practice provides a favorable cost-benefit ratio to those who aspire to maximize post-exercise muscular adaptations.

KEY POINT

Numerous studies support the efficacy of nutrient timing for acutely increasing muscle protein synthesis following a resistance training bout, but research has failed to demonstrate that protein timing has a long-term effect on muscle hypertrophy. However, given that there are no discernable detriments to the practice and given that it may be of benefit, those who aspire to maximize hypertrophy should consume protein soon after finishing a resistance training bout.

Effect of Post-exercise Carbohydrate on Hypertrophy

The inclusion of carbohydrate in post-workout nutrition intake is often claimed to be synergistic to protein consumption with respect to promoting a hypertrophic response (58). This assertion is primarily based on theorized anabolic actions of carbohydrate-mediated insulin release. However, although insulin has known anabolic properties (17, 40), emerging research shows that the hormone has a permissive rather than stimulatory role in regulating protein synthesis (106). Its secretion has little impact on post-exercise anabolism at physiological levels (48), although evidence suggests a threshold below which plasma insulin levels cause a refractory response of muscle protein synthesis to the stimulatory effect of resistance training (68). Importantly, studies have failed to show additive effects of carbohydrate on enhancing a favorable post-exercise muscle protein balance when combined with amino acid provision (45, 70, 132).

The principal effects of insulin on lean body mass are related to its role in reducing muscle catabolism (30, 43, 56, 66). Although the precise mechanisms are not well defined at this time, anticatabolic effects are believed to involve insulin-mediated phosphorylation of PI3K/Akt, which in turn blunts activation of the Forkhead family of transcription factors

(67). An inhibition of other components of the ubiquitin–proteasome pathway are also theorized to play a role in the process (48).

To take advantage of these anticatabolic properties, traditional nutrient timing lore proposes a benefit to spiking insulin levels as fast and high as possible following an exercise bout. Muscle protein breakdown is only slightly elevated immediately post-exercise and then rapidly rises thereafter (71). When in the fasted state, proteolysis is markedly increased at 195 minutes post-exercise, and protein balance remains negative (109). The extent of protein breakdown increases by up to 50% at the 3-hour mark, and heightened proteolysis can persist for up to 24 hours after an intense resistance training bout (71). Given that muscle hypertrophy represents the difference between myofibrillar protein synthesis and proteolysis, a decrease in protein breakdown would conceivably enhance the accretion of contractile proteins and thus facilitate hypertrophy.

Although the concept of spiking insulin is logical in theory, the need to do so post-exercise ultimately depends on when food was consumed pre-exercise. The impact of insulin on net muscle protein balance plateaus at 3 to 4 times fasting levels (a range of approximately 15 to 30 mU/L) (48, 113). Typical mixed meals achieve this effect 1 to 2 hours after consumption, and levels remain elevated for 3 to 6 hours (or more) depending on the size of the meal. For example, a solid meal of 75 g carbohydrate, 37 g protein, and 17 g fat raises insulin concentrations 3-fold over fasting conditions within a half hour after consumption and increases to 5-fold after 1 hour; at the 5-hour mark, levels remain double those seen during fasting (25). Hence, the need to rapidly reverse catabolic processes is relevant only in the absence of pre-workout nutrient provision.

KEY POINT

There is no need to spike insulin post-exercise via carbohydrate consumption with the goal of hypertrophy if exercise was not performed in a fasting state. The need to quickly replenish glycogen is only relevant for those who perform 2-a-day split resistance training bouts (i.e., morning and evening) in which the same muscles are worked during the respective sessions.

It also should be noted that amino acids are highly insulinemic. A 45 g dose of whey isolate produces insulin levels sufficient to maximize net muscle protein balance (15 to 30 mU/L) (110). Once this physiological threshold is attained via amino acid consumption, adding carbohydrate to the mix to further stimulate elevations in insulin is moot with respect to hypertrophic adaptations (48, 51, 132).

There is evidence that consuming carbohydrate immediately after exercise significantly increases the rate of muscle glycogen repletion; delaying intake by just 2 hours decreases the rate of resynthesis by as much as 50% (59). This is due to the potentiating effect of exercise on insulin-stimulated glucose uptake, which shows a strong positive correlation to the magnitude of glycogen use during the bout (116). Mechanisms responsible for this phenomenon include heightened translocation of the glucose transporter type 4 (GLUT4) protein responsible for facilitating entry of glucose into muscle (32, 63) and an increase in the activity of glycogen synthase—the principal enzyme involved in promoting glycogen storage (98). In combination, these factors expedite the uptake of glucose after exercise, accelerating the rate of glycogen replenishment.

Glycogen is considered critical to the performance of hypertrophy-type protocols (72). MacDougall and colleagues (82) found that 3 sets of elbow flexion exercises at 80% of 1RM performed to muscular failure decreased mixed-muscle glycogen concentration by 24%. Similar findings were reported for the vastus lateralis: 3 sets of 12RM depleted glycogen stores by approximately 26%, and 6 sets led to an approximately 38% reduction. Extrapolation of these results to a typical high-volume bodybuilding workout involving multiple exercises and sets for the same muscle group indicates that the majority of local glycogen stores are depleted during such training. Although substantial glycogen reduction occurs across all fiber types during resistance exercise, its depletion is particularly evident in Type II fibers (69), which have the greatest force-producing capacity and hypertrophic potential. Decrements in performance from glycogen depletion would conceivably impair the ability to maximize the hypertrophic response to exercise.

Despite a reliance on glycolysis during resistance training, the practical importance of rapid glycogen replenishment is questionable for the majority of lifters. Even if glycogen is completely depleted during exercise, full replenishment of these stores is accomplished within 24 hours regardless of whether carbohydrate intake is delayed post-workout (41, 102). Thus, the need to quickly replenish glycogen is only relevant for those who perform 2-a-day split resistance training bouts (i.e., morning and evening) in which the same muscles are worked during the respective sessions (6). The rate of glycogen repletion is not a limiting factor in those who consume sufficient carbohydrate over the course of a day. From a muscle-building standpoint, the focus should be directed at meeting the daily carbohydrate requirement as opposed to worrying about timing issues.

In terms of nutrient timing, there is compelling evidence that the body is primed for anabolism following intense exercise. Muscles become sensitized to nutrient intake so that muscle protein synthesis is blunted until amino acids are consumed. However, the body of research suggests that the anabolic window of opportunity is considerably larger than the 1-hour post-workout period often cited in the literature. Moreover, there is evidence that the relevance of the post-workout window of opportunity is dependent on the timing of the pre-workout meal. In a proof-of-principle study, my lab randomized resistance-trained men to consume a supplement containing 25 g of protein either immediately before performance of total-body resistance exercise or immediately after the workout (125). After 10 weeks, both groups experienced similar changes in fat-free mass and muscle thickness measures regardless of the timing of protein consumption. The findings indicate that consuming a protein-rich meal before exercise increases the duration of the post-exercise anabolic window; alternatively, if training is undertaken in a fasted state, it becomes increasingly important to consume protein soon after the bout to promote anabolism.

The practical application of nutrient timing should therefore be considered for the entire peri-workout period (before, during, and after workout). Although research is somewhat equivocal, it seems prudent to consume high-quality protein (at a dose of approximately 0.4 to 0.5 g/kg of lean body mass) both pre- and post-exercise within about 4 to 6 hours of each other depending on meal size. For those who train partially or fully fasted, on the other hand, the need to ingest protein immediately post-workout is of greater consequence.

PRACTICAL APPLICATIONS

NUTRIENT TIMING GUIDELINES

It is important to consume high-quality protein (at a dose of approximately 0.4 to 0.5 g/kg of lean body mass) both pre- and post-exercise within about 4 to 6 hours of each other depending on meal size. Those who resistance train partially or fully fasted should consume protein (at a dose of approximately 0.4 to 0.5 g/kg of lean body mass) as quickly as possible post-workout, preferably within 45 minutes of the bout. Those who perform 2-a-day workouts (morning and evening bouts in the same day) should consume carbohydrate (at a dose of approximately 1.0 to 1.5 g/kg of lean body mass) within 1-hour post-workout.

TAKE-HOME POINTS

- A positive energy balance is necessary for maximizing the hypertrophic response to resistance training. Those with limited resistance-training experience can benefit from a higher energy surplus without incurring significant adipose accretion. Alternatively, well-trained individuals require a smaller surplus (≤500 kcal/day) to prevent unwanted body fat deposition.
- Those seeking to maximize hypertrophy should consume at least 1.6 g/kg/day of protein, and perhaps as much as 2.2 g/kg/day. Qualitative factors are not an issue for those eating a meat-based diet. Vegans must be cognizant of eating a variety of protein sources over time so that they get sufficient quantities of the full complement of EAAs.
- Carbohydrate intake should be at least 3 g/kg/day to ensure that glycogen stores are fully stocked. Higher carbohydrate intakes may enhance performance and anabolism, but this may be specific to the individual.
- Dietary fat should comprise the balance of nutrient intake after setting protein and carbohydrate amounts. People should focus on obtaining a majority of fat from unsaturated sources.
- To maximize anabolism, a protein intake of 0.4 to 0.55 g/kg/meal should be spread across a minimum of four meals to reach a total of 1.6 to 2.2 g/kg/day.
- Nutrient timing around the exercise bout should be considered in the context of the peri-workout period. It seems prudent to consume high-quality protein (at a dose of approximately 0.4 to 0.5 g/kg of lean body mass) both pre- and post-exercise within about 4 to 6 hours of each other, depending on meal size. Those who train partially or fully fasted should consume protein as quickly as possible post-workout.